

Project Description – Project Proposals

Jeannette Stark, TU Dresden

Contextual Cues for Habit-Based Behavior Change in Digital Health Interventions (CuB)

Project Description

1 Starting Point

1.1 Introduction and Problem Statement

Noncommunicable diseases (NCDs) represent one of the most critical contemporary health challenges [1]. Cardiovascular diseases, cancers, chronic respiratory diseases, and diabetes account for roughly 74% of all deaths worldwide and impose substantial individual, societal, and economic costs [2]. Every day, health-related behaviors play a central role in the onset and progression of NCDs [3]. Nutrition influences obesity and diabetes risk [4], and insufficient physical activity contributes to cardiovascular disease [5], [6], [7], underscoring the preventive potential of sustained, health-promoting behaviors [8]. This project addresses the challenge of realizing this preventive potential at scale by examining the sustained formation of health-promoting behavior, thereby contributing to reducing NCD risk.

So far, however, sustaining health-promoting behaviors in everyday life has proven difficult. This challenge is particularly evident in digital health interventions (DHIs). DHIs increasingly support NCD prevention and management [9] by translating medical knowledge and behavioral recommendations into digital functionalities such as reminders, feedback loops, gamification elements, and social interaction [10]. Empirical studies report positive short-term effects during periods of active DHI use across domains such as diabetes prevention ([11], [12]), smoking cessation ([13]), weight management ([14], [15], [16]), and cardiovascular risk reduction ([17], [18]). Yet these gains often diminish once active DHI interaction declines.

Behavioral science identifies habit formation, defined as the process by which repeated behavior becomes increasingly automatic and cue-driven, as a key mechanism for stabilizing behavior over time [19], [20], [21]. Habits form when a contextual cue is repeatedly linked to a behavior, such as going for a 20-minute walk (behavior) after dinner each evening (cue). As this cue-behavior link strengthens, the behavior requires less conscious reflection, motivation, or external prompting [22], [23]. A key strategy for supporting habit formation is the use of implementation intentions, which translate a general goal into an if-then plan by linking a behavior to a contextual cue [24], [25]. For example, a general goal such as "I aim to walk more often" can be translated into the following implementation intention: "If I finish dinner, then I will go for a 20-minute walk" [25], [26]. By repeatedly linking the same cue to the same behavior, implementation intentions strengthen the cue-behavior link and increase the likelihood that the behavior is enacted automatically when the situation arises [26]. Although this cue-based logic is highly relevant for DHIs, cue support is often reduced to generic reminders, notifications, or time-based prompts [27], [28], [29], [30], [31]. While such functionalities may support engagement during active system use, they do not create the stable cue-behavior links required for habits to persist once DHI interaction declines. As a result, behavior enactment often attenuates once active DHI use decreases, because the behavior was primarily maintained through continued system interaction rather than through habits embedded in everyday life (see fading trajectory in Figure 1).

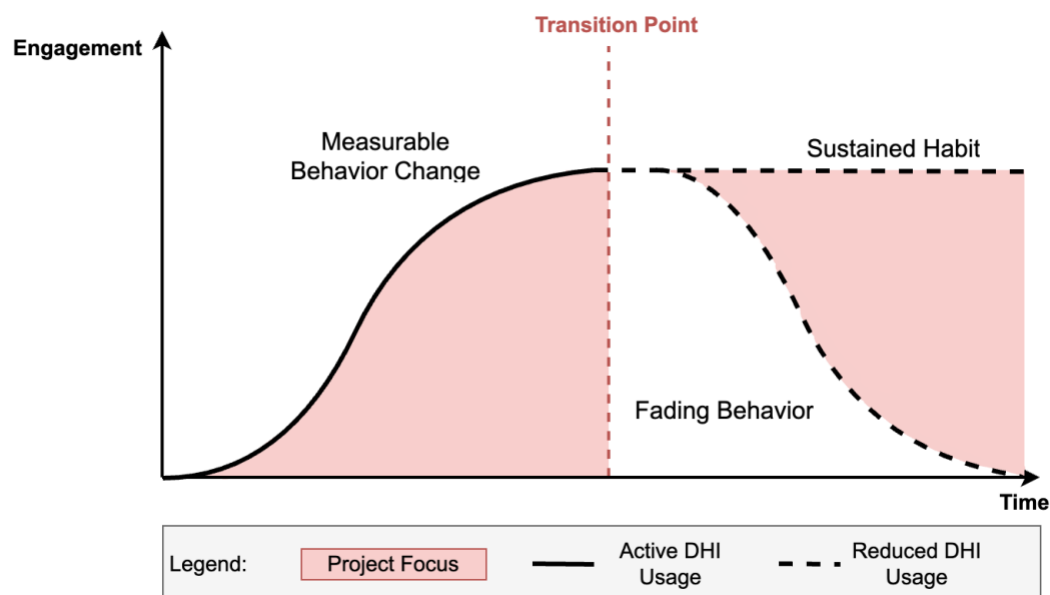


Figure 1: Stylized trajectory of DHI-supported behavior change: during active use, interventions can generate short-term behavioral gains; once DHI-support declines, behavior either stabilizes or attenuates.

To support habits that persist beyond active system use, DHIs must do more than prompt behavior through reminders or require continued system interaction. They need to help users identify effective contextual cues and link these cues to the behaviors the users aim to implement, so that the cue-behavior link strengthens until the habit becomes automatic [32]. For such support to be effective, however, DHIs require guidance on what makes a contextual cue a suitable anchor for behavior. Available evidence suggests that cues may differ in this regard. Some cues may be more reliable than others in triggering behavior because they occur more often in everyday life (cue stability), are easier to recognize when they arise (cue salience), clearly signal a particular opportunity for action (cue specificity), or better fit a person's existing routines and situations (person-cue fit). Yet the foundational knowledge required to systematically distinguish and operationalize such cues remains largely absent [33], [34], [35].

Prior research has examined individual cue properties in isolation but has not systematically investigated how cue qualities (e.g., stability, salience, specificity), cue combinations, and person-cue fit jointly relate to habit-formation outcomes [29], [36]. As a result, DHI designers lack the empirical evidence needed to select and configure contextual cues in habit-support interventions. Consequently, DHIs do not implement stable cue-behavior links that habits require to persist beyond active DHI engagement. To address this problem, this project develops a cue systematization that characterizes contextual cues along multiple dimensions and then investigates how specific dimensions within implementation intentions influence habit formation in a pre-registered longitudinal field study. The project therefore addresses the following research question:

How can contextual cues be systematized for habit formation and which cue dimensions influence the development of sustained health-promoting habits?

1.2 Habit Formation: Implementation Intentions and Situational Cues

The following section summarizes prior research on habit formation, represented by the grey elements in Figure 2, to establish the conceptual foundation for the project's research problem. Prior research defines habit formation as the process by which a behavior, repeated in stable contexts, becomes increasingly automatic [19], [22], [37]. The habit formation process is characterized by the development of automaticity, which is commonly assessed as a psychological outcome using the Self-Report Habit Index (SRHI). The SRHI captures whether a behavior is experienced as automatic, effortless, and cue-triggered [37]. Higher SRHI scores indicate stronger cue-behavior links and thus stronger habits, understood as behaviors that are increasingly governed by contextual cues and less dependent on conscious deliberation.

Beyond psychological automaticity, prior research also assesses habits through behavioral outcomes. As automaticity increases, behavior is expected to become more consistently executed over time and less dependent on continued intervention support, which is reflected in increased behavior enactment and lower dropout in Figure 2. Together, SRHI, behavior enactment, and dropout capture whether a behavior is not merely repeated but increasingly stabilized as a habit [37].

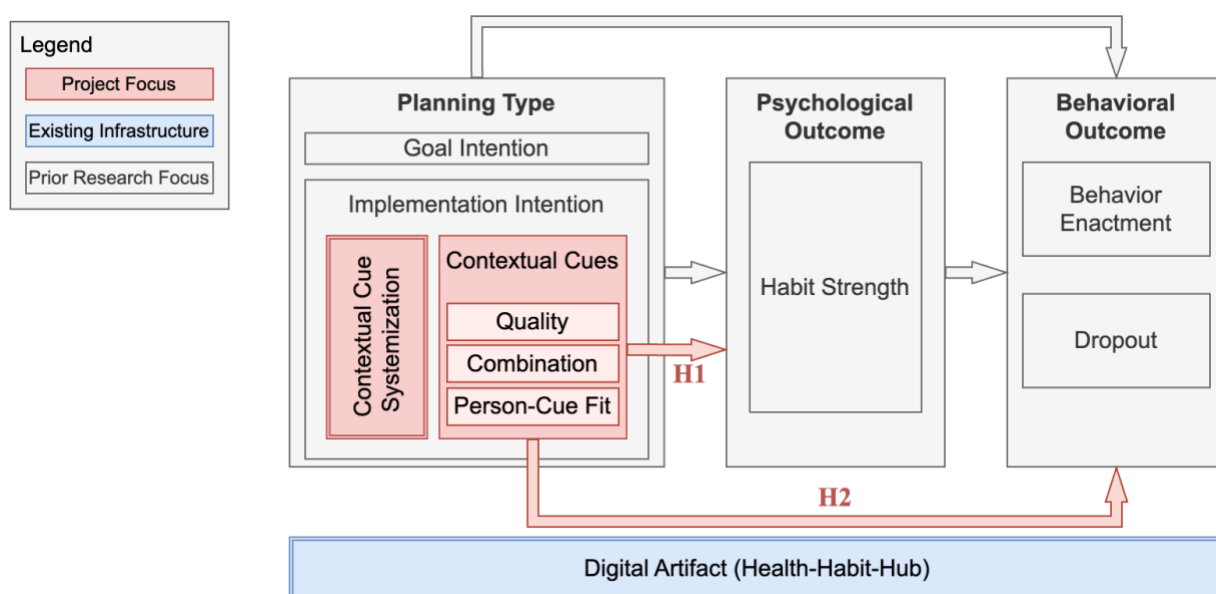


Figure 2: Conceptual framework of the project. Grey elements depict mechanisms established in prior research. Red elements highlight the project's focus: the cue configuration specified within an implementation intention and its effects on habit strength (H1) and behavioral outcomes (H2). Blue denotes the Health-Habit-Hub, the existing digital infrastructure through which all cue configurations are instantiated.

As habit formation depends on repeated behavior in response to recurring contextual cues, habit research has emphasized implementation intentions as a key strategy for supporting habit formation beyond goal intentions [24], [25]. By linking a behavior to a contextual cue, implementation intentions make the cue more accessible and the behavior easier to recall when the situation arises, leading to increased habit strength as measured with SRHI scores [19], [23]. When habits become strong, behavior enactment increases while dropout decreases [37], [38], [39]. Both the effect of implementation intentions on habit strength and their effect on behavior enactment and dropout have been shown across domains such as workplace recycling, flossing, and exercise [40], [41], [42].

While the effects of implementation intentions on habit strength and the link between habit strength and sustained behavior are well established (see grey elements in Figure 2), the cue component of implementation intentions remains less understood. Prior research shows that specifying a contextual cue is important [24], [25], but it provides limited guidance on selecting and integrating cues into implementation intentions. Studies have varied the behavior extensively but have examined contextual cues only narrowly, typically focusing on a specific time of day, a single location, or a single routine. Moreover, these cues are often chosen pragmatically rather than systematically [10], [27], [28].

However, available evidence suggests that not all contextual cues are equally effective. Some cues may be more reliable than others in triggering the behavior. As an initial conceptual approximation, cue reliability can be approached through three aspects: cue quality, cue combination, and person-cue fit. First, cue quality concerns how stable, salient, and specific a cue is in everyday life [26]. Stability refers to how consistently a cue occurs across days and weeks [43]. Salience describes how easily a cue is noticed in the moment when the behavior should be triggered [22], [44]. Specificity refers to how uniquely the cue signals a particular opportunity for action [45]. These cue qualities may vary independently: a cue can be stable but low in salience, or highly salient but low in specificity. For example, "11 o'clock each day" is highly stable and specific but low in salience and may go unnoticed without an external signal, whereas "when I feel like snacking" may be salient in the moment but is low in stability and specificity, making it harder to detect reliably and link to a consistent behavioral response. Second, cue combinations concern how multiple cues are linked into an implementation intention. A single cue, such as "after lunch," may provide a weaker situational anchor than a combination, such as "after lunch at work on weekdays," because the latter defines a more specific and recurring context. Third, initial evidence also points to the relevance of a person-cue fit. Cues may be more effective when they align with a person's everyday routines, situation, and needs rather than being assigned generically [46]. Together, cue quality, cue combination, and person-cue fit provide an initial conceptual approximation of what this project refers to as cue systematization, while acknowledging that additional cue properties and relationships need to be explored and added.

While initial studies have approached contextual cues by examining individual aspects such as cue selection [35] or consistency with a single cue [34], a systematic investigation of contextual cues under controlled, ecologically valid conditions remains limited, although it has been identified as a research priority in habit-based behavior change [36]. This research gap creates problems across habit research and digital health. In habit science, the absence of a validated cue systematization renders it difficult to design controlled experiments that systematically vary and compare cue qualities, multi-cue configurations, and forms of person-cue fit. As a result, evidence on which cues best support automaticity remains fragmented [10], [29], [36]. In digital health research, the same problem becomes a design problem. DHIs need to translate contextual cues into digital intervention components such as sensors, contextual prompts, or adaptive support. Without a cue systematization and the empirical evidence it enables, these design choices are being guided by convention rather than evidence [27], [30], [32].

This project aims to systematize contextual cues for digital habit-support interventions (see red elements in Figure 2). Once established, this systematization provides the basis for a longitudinal field study investigating how cue dimensions influence habit formation at the psychological and

behavioral level. Accordingly, we derive two confirmatory hypotheses (H1 and H2a) and one exploratory hypothesis (H2b), summarized in Table 1.

Psychological Outcome	Behavioral Outcome
H1: Differences in cue dimensions as part of the implementation intention influence habit strength.	H2a: Differences in cue dimensions as part of the implementation intention influence behavior enactment. H2b: Differences in cue dimensions as part of the implementation intention influence dropout.

Table 1. The three hypotheses tested by the project correspond to the project-focus relationships (red arrows) in Figure 2. H1 (Psychological Outcome) addresses habit strength; H2 (Behavioral Outcome) addresses behavior enactment and dropout.

All three hypotheses are deliberately framed in non-directional terms. Prior evidence on cue effectiveness remains fragmented [33], [34], [35], and several plausible effect patterns are consistent with it. Richer cue configurations may strengthen the cue–behavior link by providing a more reliable situational anchor, while sparser configurations may prove more robust when they correspond to a single, highly stable context. Person-cue fit may amplify these effects when cues integrate with existing routines or attenuate them when self-selected cues trade theoretical quality for personal familiarity. H1 and H2a are therefore pre-registered in terms of which cue dimensions are varied and which outcomes are tested, with two-tailed tests to remain agnostic about direction. H2b uses the same planned contrast structure for estimating cue effects on dropout, but is treated as exploratory because realistic dropout hazard ratios ($HR \approx 0.80\text{--}0.90$) require analyzable samples of approximately 3,400 to 14,700. For H1 and H2a, the project pre-registers three planned, between-participant contrasts corresponding to the three cue dimensions: (i) high vs. low cue quality (200 vs. 200 participants), (ii) multi-cue vs. single-cue combination (300 vs. 300 participants), and (iii) self-selected vs. researcher-assigned cues (200 vs. 400 participants). For H1 and H2a separately, family-wise Type I error is controlled at $\alpha = 0.05$ by Bonferroni correction across the three contrasts (per-contrast threshold $\alpha = 0.0167$). For H2b, the same contrasts are reported as exploratory hazard-ratio estimates with 95% confidence intervals.

1.3 Our Prior Work

The proposed project builds on prior work conducted within the Research Group Digital Health at TU Dresden. This work provides both the conceptual foundation and the technical infrastructure required for the proposed project. Conceptually, our prior research has shown that contextual cues are a central unresolved issue in digital habit formation. Technically, the group has developed the Health-Habit-Hub, a digital research platform that can be extended to instantiate and test cue-based implementation intentions in a longitudinal field study.

In a first line of work, we analyzed the digital habit app market and consolidated a systematic overview of design principles spanning the habit formation process [10], [47]. This work identified design principles directly relevant to contextual cueing and implementation intentions, including contextual specification and variables, reminders to reinforce implementation intentions, tapered reminders (e.g., of implementation intentions), and fading. This overview of design principles relevant for habit formation provides the conceptual starting point for WP A, which focuses

specifically on the cue-related part of this prior design-principle framework established in [10]. Building on this work, a follow-up analysis examined 36 contemporary habit apps and extracted 585 behavior recommendations from the apps [48]. The study showed that habit apps commonly address multiple lifestyle domains and that apps covering a broader range of domains tend to offer more habit-forming functionalities. At the same time, user ratings did not simply increase with the number of supported behaviors and functionalities, indicating that functional breadth alone does not explain user satisfaction. For the present project, the analysis provides two operational insights: First, physical activity emerged as the most frequently represented and strongly interconnected behavioral domain in current apps, with frequent co-occurrences with nutrition, leisure activities, and sleep. Physical activity is therefore a well-justified target domain for the field study in WP C, because it is both widely represented in existing app practice and appears to offer spillover potential into other health-relevant lifestyle domains. Second, the study shows that current apps already provide rich behavior recommendations and habit-support functionalities but also reveals that functionalities remain organized around app-internal design patterns rather than around evidence on how contextual cues should be selected and configured for habit formation.

A complementary investigation compared how behavioral scientists and end users evaluate the relevance of previously identified design principles [30]. The study revealed a systematic perception gap. Users assigned lower importance to contextual cues than scientists, although contextual cues are central to habit formation. Combined with the finding that current apps often implement contextual cues as simple time-based reminders rather than richer contextual embedding, this finding shows that cue design in DHIs cannot be derived from user preference or current app practice alone. Instead, cue-based intervention components require scientific grounding, while user input should inform personalization within evidence-based defaults. Developing this evidence base for contextual cues is the contribution of the present project.

Generating this evidence base requires a digital research instrument that can implement controlled cue interventions and observe participants over several weeks. The Research Group Digital Health has developed and deployed such an instrument. The Health-Habit-Hub combines a participant-facing habit app with a researcher-side study environment. Its technical foundations build on prior design science work from vCare, an EU-funded Horizon 2020 project on virtual coaching for rehabilitation in older adults [49]. This work provided design knowledge on modular behavior tracking, goal specification, and adaptive coaching, which was later transferred and evolved into the Health-Habit-Hub [50]. Participants use an app to set health goals, log daily behaviors, answer questionnaires, and formulate implementation intentions with self-selected or pre-selected cues. Researchers use a separate web portal to configure studies, define experimental conditions, enroll participants, monitor SRHI and behavioral trajectories, and export data for analysis. Each recorded habit is mapped to the Behaviour Change Intervention Ontology (BCIO) [51]. This mapping aligns the platform's terms used to describe behaviors with the international evidence base and makes participant data comparable across studies and prior research syntheses. Central to this proposed project, the platform's implementation-intention module will incorporate the cue configurations produced by WP A as configurable study conditions. The field study in WP C can therefore be implemented largely through study configuration rather than new app development. Beyond the present project, the Health-Habit-Hub is designed as an open research infrastructure. Other research teams will be able to deploy the platform, configure independent habit-formation studies, and test additional cue dimensions

or intervention components. In this way, the platform is not limited to the proposed project but becomes a shared instrument for cumulative habit research, enabling future studies to build on comparable data structures, measurement routines, and intervention configurations.

The platform's role as a research instrument has already been investigated in a first empirical pilot study [52]. This study used the Health-Habit-Hub to compare just-in-time and fading reminder strategies. The results indicate that a gradual transition between reminder modes can support sustained automaticity and that not only cue content, but also the temporal structure of cue delivery influences habit formation. The pilot also confirmed the platform's stability across multi-week observation windows [52]. The Health-Habit-Hub is therefore ready to host the longitudinal field study in WP C with limited additional build-up time.

2 Objectives and work program

2.1 Anticipated total duration of the project

The project is planned for a total duration of 36 months.

2.2 Objectives

The objectives of the proposed research project are twofold. First, building on prior research and preliminary theoretical groundwork, we develop a sound, integrative, and multi-dimensional systematization of contextual cues, characterizing them along three dimensions: cue quality, cue combination, and amenability to person-cue fit. The resulting systematization enables intervention designers and researchers to systematically select, combine, and compare contextual cues in a structured manner. It thereby closes the conceptual gap identified above and is the prerequisite for Objective 2. Without the systematization, the conditions of the field study cannot be selected on a principled, theoretically grounded basis. Second, building on this systematization, we conduct a pre-registered longitudinal field study, examining two crossed between-participants factors—cue combination (single vs. multi-cue) and cue source (low-quality pre-rated, high-quality pre-rated, self-selected)—yielding $2 \times 3 = 6$ conditions; person-cue fit is operationalised as the self-selected level of cue source. The study generates the first causal evidence on how cue configurations within implementation intentions influence habit formation. Taken together, the systematization (Objective 1) and the field study (Objective 2) close the conceptual and empirical gaps identified above and translate into actionable design knowledge for DHIs. Beyond the immediate empirical yield, the systematization itself enables cumulative research: future studies can populate additional regions of the solution space and aggregate findings on cue effectiveness in a comparable manner across studies.

2.3 Work program incl. proposed research methods

Figure 3 outlines the project plan, including work packages (WPs), subtasks, and the project timeline. The work program is structured into three core WPs (A, B, C) plus one cross-cutting WP (D). WP A develops a systematization of contextual cues (Objective 1) and provides the conceptual foundation for the project. WP B translates this foundation into an operational study setup and pretests the study platform. WP C conducts the longitudinal field study, including preparation, data collection, and analysis (Objective 2). WP D accompanies the core WPs by consolidating results and supporting publication, open-science documentation and disseminating the results across the project lifecycle. The project plan is shown in Figure 3.

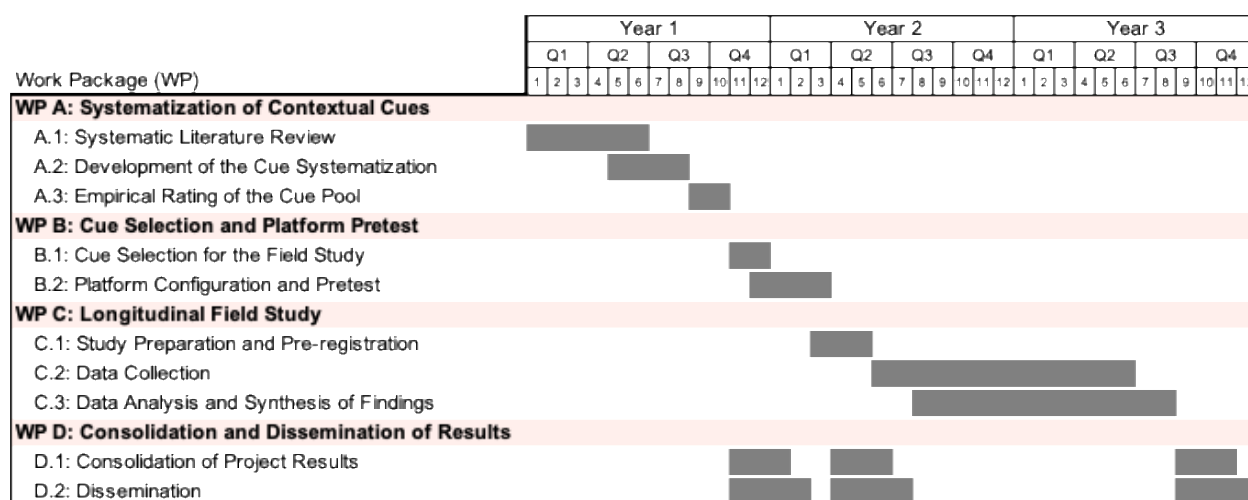


Figure 3. Project plan with work packages and subtasks over 36 months.

WP A: Systematization of Contextual Cues

Addressing the conceptual gap identified in Section 1.2, WP A develops an integrative, multidimensional systematization of contextual cues. It combines a top-down systematic literature review with a bottom-up empirical rating of the assembled contextual cue pool. The resulting systematization provides the operational foundation for WPs B and C, enabling intervention designers and researchers to systematically select, combine, and compare cues.

WP A.1: Systematic Literature Review

Result: Synthesized evidence base on contextual cues across NCD-relevant behavioral domains.

The project begins with a PRISMA-compliant systematic literature review following established guidelines [53]. The review synthesizes evidence on contextual cues in digital habit-support interventions including cue types, operationalizations, behavioral domains, cue combinations, and reported evidence on how different cue types and cue properties affect habit formation outcomes. To ensure comprehensiveness, the search covers leading IS, behavioral science, and digital health outlets via standard scientific databases. Particular attention is given to cue quality (specifically stability, salience, and specificity), as well as cue combination and amenability to person-cue fit.

WP A.2: Development of the Cue Systematization

Result: Multi-dimensional cue systematization of contextual cues.

The review findings feed into the systematization, which is developed following the iterative, hybrid method of Nickerson, Varshney & Muntermann [54] and Feine et al. [55]. We combine conceptual-to-empirical (top-down) and empirical-to-conceptual (bottom-up) cycles. We aim to classify cues by application domain, cue combination, cue quality (stability, salience, specificity), and amenability to person-cue fit, producing a structured systematization of contextual cues. To support semi-automated cue classification and ontology alignment with the BCIO, locally executed language models may be used to suggest initial categorizations of cue descriptions, with all categorizations subject to expert verification and without transfer of research data to external services. The systematization is theoretically grounded and explicitly extensible; comprehensiveness is a goal to be evaluated through the iterative cycles in WP A.2 and A.3. Subsequent studies, including the field study in WP C, populate specific regions of the systematization, and future work can extend coverage cumulatively.

WP A.3: Empirical Rating of the Cue Pool

Result: Pre-rated cue pool characterized on stability, salience, and specificity, combination and amenability to person-cue fit.

To enable principled cue selection in WP B, the systematization includes a pre-rated pool of candidate cues. Cues identified in the review are rated on stability, salience, specificity, and amenability to person-cue fit through a structured expert-rating procedure, ensuring that the ratings reflect both theoretical criteria and lived experience. The pre-rated pool is the direct input to the cue-selection step in WP B.1.

WP B: Cue Selection and Platform Pretest

WP B prepares the longitudinal field study by (1) selecting the specific cues used in the study conditions and (2) configuring and pretesting the study platform.

WP B.1: Cue Selection for the Field Study

Result: Finalized set of cues for the six study conditions, drawn from the pre-rated pool.

The full systematization of contextual cues cannot be exhaustively explored in a single study. WP B.1 therefore selects a deliberate part that spans the project's cue-quality dimensions: a set of low-quality cues (low ratings on stability, salience, and/or specificity) and a set of high-quality cues (high ratings on all three dimensions). The selection is guided by an expert panel in a focus-group format, drawing on researchers in habit science and digital health intervention design to ensure that the chosen cues represent meaningful and ecologically plausible quality contrasts. All selected cues are drawn from the pre-rated pool produced in WP A.3, ensuring that the study conditions in WP C operationalize theoretically grounded cue-quality contrasts. The part spanned by this selection is intentionally narrow; populating further regions of the systematization is left to future replication and extension work, for which the systematization itself is designed.

WP B.2: Platform Configuration and Pretest

Result: Platform validated for the field study, including measurement instruments, randomization logic, and system stability.

The Health-Habit-Hub, the existing experimental platform of the Research Group Digital Health at TU Dresden, serves as the methodological backbone of the field study. It supports configurable digital habit interventions through goal setting, behavioral tracking, SRHI administration, and a cue-specification module. In addition to the existing web-based platform, the Health-Habit-Hub will be deployed as a native iOS and Android application to improve participant accessibility and enable context-sensitive cue delivery through mobile device capabilities. WP B.2 configures the platform for the study by instantiating the six study conditions, embedding adaptive SRHI assessments, and pre-loading the cue selections from WP B.1. It then conducts a small-scale pretest with 20 to 30 participants over up to two months to validate measurement instruments, randomization logic, onboarding flow, and system stability prior to full deployment in WP C.

WP C: Longitudinal Field Study

WP C conducts a pre-registered longitudinal field study using a 2 × 3 between-participants design.

WP C.1: Study Preparation and Pre-registration

Result: Pre-registered analysis plan and study protocol on the Open Science Framework.

Prior to data collection, the complete confirmatory analysis plan is submitted to the Open Science Framework. The pre-registration covers the hypotheses, sampling logic, primary and secondary outcomes, model specifications, and robustness checks. It also includes a simulation-based

power analysis that validates the sample-size target for the confirmatory SRHI and daily-enactment outcomes, and an event-count sensitivity analysis showing that dropout effects of realistic magnitude are not confirmatorily powered and will therefore be analysed exploratorily. Pre-registration ensures strict separation between confirmatory and exploratory analyses, in line with current open-science standards.

The sample-size rationale is anchored in the minimum detectable effect for H1, the project's primary psychological outcome. We target an effect of approximately $d = 0.35$ on the SRHI asymptote, defined as the level of habit strength to which participants converge over the observation period under a nonlinear asymptotic growth model. Given the assumed between-participant asymptote SD of 0.85, this corresponds to a raw difference of approximately $\Delta = 0.30$ on the 1–7 SRHI scale between high- and low-quality cue conditions. This target sits at the upper end of the plausible between-cue range in the existing literature, which spans from null direct cue-type contrasts ([33]) to a cue-stability effect of approximately $d = 0.45$ ([43]). For reference, the meta-analytic average effect of habit-formation interventions versus control is in the same order of magnitude ([56], $SMD = 0.31$). Effects below approximately $d = 0.25$ correspond, under the simulated between-participant asymptote variance, to raw differences of approximately 0.20–0.25 points on the 1–7 SRHI scale. We treat this range as the lower bound of practical interest because such differences are unlikely to yield clearly differentiable design recommendations for DHI developers.

A Monte Carlo power simulation from a nonlinear asymptotic growth model of weekly SRHI trajectories, with participant-level random effects on asymptote, baseline, and rate constant, shows that detecting an effect of $d = 0.35$ on the asymptote with 80% power at Bonferroni-corrected $\alpha = 0.0167$ requires at least 90 analyzable participants per cell for the high- versus low-quality cue contrast. The simulation used 1,000 iterations per scenario and analysed each simulated dataset using a conservative two-stage approximation to the planned nonlinear mixed-effects model. Participant-level asymptotes were estimated from individual trajectories by closed-form OLS on the asymptotic regression and then compared across conditions using a Welch two-sample test. At the planned analyzable sample of 100 participants per cell, the achieved simulated power is 0.885. We round up to 100 participants per cell because this matches the per-arm samples used in comparable longitudinal habit-formation studies that have modelled individual trajectories analytically (Lally et al. [57], $N = 96$; Kaushal & Rhodes [58], $N = 111$; Keller et al. [33], 96 per arm; Stadler, Oettingen & Gollwitzer [59], approximately 128 per arm), providing convergent evidence that approximately 100 per cell is the right granularity for the planned trajectory analysis. The analyzable sample target is therefore $N = 600$, corresponding to six cells $\times$ 100 participants. To protect the analyzable target against attrition, we inflate the recruitment target using an assumed 40% nine-month attrition rate, treated as a conservative upper-bound estimate based on app-based intervention evidence [60]. This yields an enrollment target of approximately 165 participants per cell, or $N \approx 1,000$ in total.

For H2a, daily target-behavior enactment is modelled as a binary repeated outcome comprising 273 daily observations per participant (7 days $\times$ 39 weeks), using a participant-level logistic random-intercept data-generating model. Under the pre-specified assumptions, the reference arm has a baseline daily enactment probability of 0.30. Participant-level heterogeneity is set to $\sigma_u = 0.90$ on the log-odds scale, corresponding to an intraclass correlation of approximately $ICC = 0.20$. The cue-quality effect is specified as a subject-specific conditional odds ratio of $OR = 1.35$,

corresponding to the fixed-effect parameter in the random-intercept model. This translates to an approximate population-averaged marginal OR of 1.30. Under these assumptions, an analogous Monte Carlo simulation yields approximately 82% power at $N = 600$ analyzable participants using the same Bonferroni-corrected threshold of $\alpha = 0.0167$. The H2a analysis is intentionally conservative: each participant's 273 daily binary outcomes are reduced to a continuity-corrected empirical logit, and the high- versus low-quality cue groups are then compared using a Welch two-sample test, which loses precision relative to a full generalized linear mixed-effects model (`lme4::glmer`). The minimum detectable subject-specific OR at 80% power at $n = 100$ per cell is $OR = 1.35$ from the simulation grid; the design is not powered for subject-specific ORs below 1.35 at this sample size.

For H2b, dropout is treated as a time-to-event outcome modelled under an exponential (constant-hazard) survival assumption. Because realistic between-cue hazard ratios in this domain are expected to lie in the $HR \approx 0.80\text{--}0.90$ range, we apply Schoenfeld's (1981) event-count approximation for a two-group log-rank test, assuming 40% nine-month cumulative dropout in the reference arm (low-quality cue cells) and equal arm allocation [60], [61]. Under these assumptions, the required analyzable sample sizes in the full six-cell design are approximately 3,432 for $HR = 0.80$, 6,324 for $HR = 0.85$, and 14,726 for $HR = 0.90$ —all far beyond the project scope. The corresponding confirmatory power at the project's analyzable target of $N = 600$ is 0.149 for $HR = 0.80$, 0.081 for $HR = 0.85$, and 0.041 for $HR = 0.90$. The minimum detectable hazard ratio at 80% power at $N = 600$ is approximately $HR \approx 0.56$, corresponding to a 44% reduction in the dropout hazard—an implausibly large effect relative to the literature range. H2b is therefore pre-registered as exploratory: the project will estimate the dropout hazard ratio and its 95% confidence interval as descriptive evidence to inform the design of an adequately powered follow-up trial.

Null findings on planned confirmatory contrasts at Bonferroni-corrected $\alpha = 0.0167$ will be accompanied by equivalence tests using bounds of $\pm d = 0.25$ where the outcome scale permits a standardized mean-difference interpretation. These tests will distinguish statistically inconclusive results from effects that are sufficiently small to be considered practically negligible under the project's stated minimally interesting effect criterion.

WP C.2: Data Collection

Result: Longitudinal dataset comprising baseline data, daily behavioral logs, repeated SRHI trajectories, and dropout records for the full enrolled sample.

Recruitment proceeds through the institutional network of TU Dresden and the Research Group Digital Health, including university mailing lists, partnered research and community networks, and recruitment channels established through the group's prior digital health studies. Eligible adults (aged 18 or older, with sufficient language fluency) adopt a daily physical-activity behavior of approximately 5-20 minutes' duration as the focus of habit formation. The behavior is chosen by the participant from a predefined menu of methodologically comparable activities (e.g., walking, light jogging, cycling, structured calisthenics, yoga), ensuring that cue effects can be compared across conditions while preserving the ecological validity that personally chosen behaviors provide. Participants are then randomly assigned to one of six study conditions via stratified randomization (stratified on baseline SRHI and selected target-behavior type). The design crosses two between-participants factors: cue combination (single vs. multi-cue) and cue source (low-quality pre-rated, high-quality pre-rated, self-selected)—yielding $2 \times 3 = 6$ conditions; person-

cue fit is operationalised as the third level of cue source, in which participants select their own cue rather than receiving an assigned one. Table 2 specifies the six resulting conditions, with illustrative example implementation intentions for the target behavior "go for a 20-minute walk."

		Cue Combination	
		Single cue	Multi-cue
Cue Quality	Low quality cue (pre-rated)	C1: "When I have some free time in the evening, I will go for a 20-minute walk."	C2: "When I get home in the evening and have some free time, I will go for a 20-minute walk."
	High quality cue (pre-rated)	C3: "After dinner each evening, I will go for a 20-minute walk."	C4: "After dinner each evening, at home on weekdays, I will go for a 20-minute walk."
Person-Cue Fit	Self-selected personal cue	C5: "After my morning coffee, I will go for a 20-minute walk."	C6: "After my morning coffee, on workdays at home, I will go for a 20-minute walk."

Table 2. Experimental conditions. All six conditions implement an implementation intention. Conditions differ on cue combination (single vs. multi-cue) and cue source (low-quality pre-rated / high-quality pre-rated / self-selected). Low- and high-quality cues are drawn from the pre-rated pool produced in WP A; self-selected cues are chosen freely by participants from the WP-A solution space.

Building on the rolling-enrollment design used by Lally et al. [57], the study replaces fixed assessment windows with continuous monitoring across each participant's individual observation period. Once data collection begins, participants can enroll at any time. After onboarding, they are followed from their personal enrollment date onward, with daily behavioral logging and weekly SRHI assessment, until they disengage or until the project's data-collection phase ends. Although enrollment is continuous, recruitment efforts are front-loaded toward the start of the data-collection phase to maximize the observation period available for each participant and thus the trajectory information available for the analyses in WP C.3. The Health-Habit-Hub's scalable infrastructure allows later participants to join throughout the project without operational overhead, supporting both the primary sample and exploratory analyses on sustained engagement of participants who enroll later in the data-collection window.

WP C.3: Data Analysis and Synthesis of Findings

Result: Confirmatory tests of cue effects on habit strength (H1) and daily behavior enactment (H2a), plus exploratory estimates of cue effects on dropout (H2b), including effect-size estimates, robustness checks, and a synthesis of cue-specific design implications.

Once data collection in WP C.2 has reached the analysis window, we will start by running several pre-processing steps (e.g., data cleansing, outlier detection, descriptive statistics, manipulation checks on the cue configurations actually enacted by participants) to ensure that all data-analysis requirements are met. We will also examine compliance with condition assignments across the six study cells, confirming that participants enacted the cue configurations prescribed by their condition before proceeding to the confirmatory analyses. In line with established approaches to longitudinal behavior-change research [57], the data analysis then comprises three main steps. In a first step, we will focus on visualizing descriptive evidence for cue-architecture effects on SRHI trajectories and behavioral enactment across the six study conditions. Doing so will help reveal key trends and patterns within and across cells. Second, using the statistical software package R, we will rely on mixed-effects modelling to test the research hypotheses. H1 (effect of

cues on habit strength) will be analyzed using a nonlinear mixed-effects asymptotic growth model of weekly SRHI trajectories with random effects on the asymptote, baseline, and rate constant at the participant level, estimating condition effects on the participant-level asymptote. H2 is analysed in two parts. H2a is tested confirmatorily using mixed-effects logistic regression on daily target-behavior enactment. H2b is analysed exploratorily using time-to-event models across the full observation period, reporting hazard ratios and confidence intervals for the planned cue contrasts. Third, we will carry out add-on analyses to check the robustness of our estimation results (e.g., per-protocol samples, physical behavioral-domain subgroup, sensitivity analyses on missing-data assumptions, potential influences of participant sex/gender) and to rule out alternative explanations for the observed effects. A live analysis pipeline operates from month 20 onward, performing automated data extraction, cleaning, and descriptive reporting for ongoing quality assurance during data collection; it does not constitute hypothesis testing. The full confirmatory analysis is completed by month 32.

WP D: Consolidation and Dissemination of Results

WP D.1: Consolidation of Project Results

Result: Final report.

To foster the sustainability and practical applicability of the project outcomes, the results of WPs A, B, and C are consolidated at specific time periods across the project (see Figure 3). At the end of WP A, the cue systematization and pre-rated cue pool are integrated into a coherent conceptual output. At the end of WP B, the selected cue pool and platform validation findings are documented as the methodological foundation for WP C. At the end of WP C, the confirmatory results for H1 and H2a and the exploratory effect-size estimates for H2b are compiled into an empirical synthesis of cue-specific design implications. On this basis, the final project report integrates the conceptual contribution (cue systematization), the methodological contribution (study design and platform validation), and the empirical contribution (causal evidence on cue-architecture effects on habit formation) into a coherent body of project knowledge.

WP D.2: Dissemination

To make the project results publicly available, the project's accumulated body of knowledge is disseminated through three complementary channels. First, two journal manuscripts are planned as the primary scientific outputs. The cue systematization and pre-rated cue pool are planned to be submitted to a practice-oriented Jourqual B journal in the digital health or behavioral intervention design space (e.g., JMIR, BMC Digital Health) approximately three months after WP A. The field study findings are submitted to a Jourqual A or A+ journal in information systems or behavioral science (e.g., MISQ or EJIS) after the confirmatory analyses and exploratory dropout analyses in WP C are completed. The associated open cue repository, pre-registration, analysis code, and de-identified datasets are released on the Open Science Framework under a CC BY 4.0 license, ensuring permanent open access to the project's empirical and methodological outputs. Second, interim findings are presented at international conferences in information systems, digital health, and behavioral science (e.g., ICIS, ECIS, CHI) throughout the project lifecycle. Two focused dissemination points are scheduled approximately three months after WPs A and B respectively, to present the cue systematization and the pretest findings ahead of the corresponding journal submissions. Further conference presentations follow as data accumulate during WP C and as the confirmatory analyses are completed. Third, the project outcomes are translated into teaching and learning materials supporting the applicant's teaching at TU Dresden in undergraduate and graduate courses dealing with topics such as digital health interventions,

behavior change, and design science. Over the project course, selected topics from the work packages are discussed and explored together with students in the form of seminars and theses, who thus directly benefit from the project as well.

2.4 Handling of research data

Responsible data handling is a core concern in digital health research, particularly given the privacy risks that accompany the deployment of digital technologies in sensitive health contexts [62]. All research data generated in this project will be collected, stored, and managed in accordance with the DFG Guidelines on the Handling of Research Data and applicable data protection legislation, including the GDPR. A formal data protection review of the Health-Habit-Hub platform was conducted by the TU Dresden Data Protection Officer, who issued clearance on 28 March 2025 (Az. 0543-025/001), confirming no data protection objections to the conduct of research using this infrastructure. In line with the Data Protection Officer's recommendation, the implementation of the anonymization protocol will be evaluated at regular intervals throughout the study given the longitudinal scale of data collection. The project generates two categories of research data. First, the project collects behavioral and psychometric data from the longitudinal field experiment: self-reported behavioral frequency, SRHI scores administered at participant-adaptive intervals, and dropout timestamps, collected via the Health-Habit-Hub platform. Second, passively logged event data capturing participants' in-app interactions with the cue specification module—including cue type selections, cue configuration choices, and behavioral logging entries—stored in the platform's Neo4j and MongoDB backends. These data are central to the study and to characterizing the cues participants adopt in practice. All data are pseudonymized at the point of collection: participants are assigned a study ID, and any contact information retained for follow-up assessment reminders is stored separately on secured TU Dresden servers and deleted upon study completion. No personally identifiable information is retained in the research database. Data are shared only within the project's research group at TU Dresden and are not passed to any other third parties, consistent with the approved informed consent procedure and the ethics committee's guidance (SR-EK-222052025).

All analysis scripts, pre-registered analytic plans, and de-identified datasets will be versioned and archived on the Open Science Framework and mirrored on GitHub for collaborative version control. The pre-registration is submitted within WP B, ensuring strict separation between confirmatory and exploratory analyses. Following project completion, de-identified raw data and analysis code will be made openly accessible under a Creative Commons license (CC BY 4.0). Archived data will be retained for a minimum of ten years following project completion on TU Dresden's institutional servers with regular backup protocols.

2.5 Relevance of sex, gender and/or diversity in the research project

Sex, gender, and diversity are relevant to this project both in terms of research content and study administration. From a content perspective, the target behaviors investigated in this project are domains in which sex- and gender-related differences in baseline behavior, habit strength, and responsiveness to behavioral interventions are well-documented [63], [64], [65], [66]. For instance, adherence rates and dropout patterns in digital health interventions have been shown to differ by sex, as have the contextual cues that individuals regard as salient and stable in their daily lives [67], [68], [69]. Accordingly, sex and gender will be recorded as participant-level variables and included as exploratory moderators in the analytic strategy. This will allow for post-hoc assessment of whether the effects of cues on habit formation differ systematically across sex

and gender groups. Beyond sex and gender, the continuous enrollment design is sensitive to sociodemographic diversity more broadly: variation in participants' living situations, working conditions, and routine structures is not treated as noise to be controlled but as ecologically valid heterogeneity that reflects the real-world conditions under which habits must be formed. This diversity strengthens the external validity and generalizability of the project's findings. From an administrative perspective, participant recruitment imposes no restrictions on sex, gender, or any other demographic characteristic. Enrollment is open and self-directed, and the study platform and materials use gender-neutral language throughout. Care will be taken to avoid any form of discrimination against individual participants or participant groups, consistent with the ethics committee's requirements (SR-EK-222052025).

3 Project- and subject-related list of publications

- [1] WHO, "Non communicable diseases," Noncommunicable diseases (NCDs). Accessed: Mar. 24, 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/noncommunicable-diseases>
- [2] WHO, *Noncommunicable Diseases Progress Monitor 2022*, 1st ed. Geneva: World Health Organization, 2022.
- [3] A. Budreviciute *et al.*, "Management and Prevention Strategies for Non-communicable Diseases (NCDs) and Their Risk Factors," *Front. Public Health*, vol. 8, p. 574111, Nov. 2020, doi: 10.3389/fpubh.2020.574111.
- [4] A. Medina-Remón, R. Kirwan, R. M. Lamuela-Raventós, and R. Estruch, "Dietary patterns and the risk of obesity, type 2 diabetes mellitus, cardiovascular diseases, asthma, and neurodegenerative diseases," *Crit. Rev. Food Sci. Nutr.*, vol. 58, no. 2, pp. 262–296, Jan. 2018, doi: 10.1080/10408398.2016.1158690.
- [5] D. Mozaffarian, P. W. F. Wilson, and W. B. Kannel, "Beyond Established and Novel Risk Factors: Lifestyle Risk Factors for Cardiovascular Disease," *Circulation*, vol. 117, no. 23, pp. 3031–3038, Jun. 2008, doi: 10.1161/CIRCULATIONAHA.107.738732.
- [6] K. Kushlev, D. M. Drummond, and E. Diener, "Subjective Well-Being and Health Behaviors in 2.5 Million Americans," *Appl. Psychol. Health Well-Being*, vol. 12, no. 1, pp. 166–187, Mar. 2020, doi: 10.1111/aphw.12178.
- [7] C. J. Lavie, C. Ozemek, S. Carbone, P. T. Katzmarzyk, and S. N. Blair, "Sedentary Behavior, Exercise, and Cardiovascular Health," *Circ. Res.*, vol. 124, no. 5, pp. 799–815, Mar. 2019, doi: 10.1161/CIRCRESAHA.118.312669.
- [8] Z.-Y. Yang, Z. Yang, L. Zhu, and C. Qiu, "Human Behaviors Determine Health: Strategic Thoughts on the Prevention of Chronic Non-communicable Diseases in China," *Int. J. Behav. Med.*, vol. 18, no. 4, pp. 295–301, Dec. 2011, doi: 10.1007/s12529-011-9187-0.
- [9] WHO, "Advancing be he@lthy, be mobile in the Eastern Mediterranean Region: combating noncommunicable diseases through mobile technology / World Health Organization." WHO Regional Office for the Eastern Mediterranean, Cairo, 2021.
- [10] J. Stark *et al.*, "Building Habits in the Digital Age: Incorporating Psychological Needs and Knowledge from Practitioners to Inform the Design of Digital Therapeutics," in *Wirtschaftsinformatik 2023 Proceedings*, Oct. 2023. doi: https://doi.org/10.1007/978-3-031-80125-9_32.
- [11] V. Nguyen, P. Ara, D. Simmons, and U. L. Osuagwu, "The Role of Digital Health Technology Interventions in the Prevention of Type 2 Diabetes Mellitus: A Systematic Review," *Clin. Med. Insights Endocrinol. Diabetes*, vol. 17, p. 11795514241246419, Jan. 2024, doi: 10.1177/11795514241246419.
- [12] L. Van Rhoon, M. Byrne, E. Morrissey, J. Murphy, and J. McSharry, "A systematic review of the behaviour change techniques and digital features in technology-driven type 2 diabetes prevention interventions," *Digit. Health*, vol. 6, p. 2055207620914427, Jan. 2020, doi: 10.1177/2055207620914427.
- [13] S. Li, Z. Qu, Y. Li, and X. Ma, "Efficacy of e-health interventions for smoking cessation management in smokers: a systematic review and meta-analysis," *eClinicalMedicine*, vol. 68, Feb. 2024, doi: 10.1016/j.eclinm.2023.102412.
- [14] L. Roth, M. Ordnung, K. Forkmann, N. Mehl, and A. Horstmann, "A randomized-controlled trial to evaluate the app-based multimodal weight loss program *zanadio* for patients with obesity," *Obesity*, vol. 31, no. 5, pp. 1300–1310, May 2023, doi: 10.1002/oby.23744.
- [15] A. Lautenbach, M. Blüher, A. Mijuskovic, L. Jones, and F. Schirmann, "Efficacy of a digital health application for weight management in people with obesity: 6-months results from a randomized controlled trial," *Int. J. Obes.*, vol. 50, no. 2, pp. 466–473, Feb. 2026, doi: 10.1038/s41366-025-01967-3.
- [16] A. M. Beleigoli, A. Q. Andrade, A. G. Cançado, M. N. Paulo, M. D. F. H. Diniz, and A. L. Ribeiro, "Web-Based Digital Health Interventions for Weight Loss and Lifestyle Habit Changes in Overweight and Obese Adults: Systematic Review and Meta-Analysis," *J. Med. Internet Res.*, vol. 21, no. 1, p. e298, Jan. 2019, doi: 10.2196/jmir.9609.
- [17] F. T. Berhe *et al.*, "Digital versus nondigital behavioral interventions on cardiovascular risk reduction: systematic review and meta-analysis," *Ann. Behav. Med.*, vol. 59, no. 1, p. kaaf043, Jan. 2025, doi: 10.1093/abm/kaaf043.
- [18] N. Gold, A. Yau, B. Rigby, C. Dyke, E. A. Remfry, and T. Chadborn, "Effectiveness of Digital Interventions for Reducing Behavioral Risks of Cardiovascular Disease in Nonclinical Adult Populations: Systematic Review of Reviews," *J. Med. Internet Res.*, vol. 23, no. 5, p. e19688, May 2021, doi: 10.2196/19688.

- [19] B. Gardner, "A review and analysis of the use of 'habit' in understanding, predicting and influencing health-related behaviour," *Health Psychol. Rev.*, vol. 9, no. 3, pp. 277–295, Aug. 2015, doi: 10.1080/17437199.2013.876238.
- [20] W. Wood and D. T. Neal, "The habitual consumer," *J. Consum. Psychol.*, vol. 19, no. 4, pp. 579–592, Oct. 2009, doi: 10.1016/j.jcps.2009.08.003.
- [21] M. Limayem, S. G. Hirt, and C. M. K. Cheung, "How Habit Limits the Predictive Power of Intention: The Case of Information Systems Continuance1," *MIS Q.*, vol. 31, no. 4, pp. 705–737, Dec. 2007, doi: 10.2307/25148817.
- [22] B. Gardner, P. Lally, and J. Wardle, "Making health habitual: the psychology of 'habit-formation' and general practice," *Br. J. Gen. Pract.*, vol. 62, no. 605, pp. 664–666, Dec. 2012, doi: 10.3399/bjgp12X659466.
- [23] J. A. Ouellette and W. Wood, "Habit and intention in everyday life: The multiple processes by which past behavior predicts future behavior.," *Psychol. Bull.*, vol. 124, no. 1, pp. 54–74, Jul. 1998, doi: 10.1037/0033-2909.124.1.54.
- [24] P. M. Gollwitzer and P. Sheeran, "Implementation Intentions and Goal Achievement: A Meta-analysis of Effects and Processes," in *Advances in Experimental Social Psychology*, vol. 38, Academic Press, 2006, pp. 69–119. doi: 10.1016/S0065-2601(06)38002-1.
- [25] P. M. Gollwitzer, "Implementation intentions: Strong effects of simple plans.," *Am. Psychol.*, vol. 54, no. 7, pp. 493–503, Jul. 1999, doi: 10.1037/0003-066X.54.7.493.
- [26] M. Bieleke, L. Keller, and P. M. Gollwitzer, "If-then planning," *Eur. Rev. Soc. Psychol.*, vol. 32, no. 1, pp. 88–122, Jan. 2020, doi: 10.1080/10463283.2020.1808936.
- [27] K. Stawarz, A. L. Cox, and A. Blandford, "Beyond Self-Tracking and Reminders: Designing Smartphone Apps That Support Habit Formation," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, in Chi '15, vol. 2015– April. Seoul Republic of Korea: ACM, Apr. 2015, pp. 2653–2662. doi: 10.1145/2702123.2702230.
- [28] C. Pinder, J. Vermeulen, B. R. Cowan, and R. Beale, "Digital Behaviour Change Interventions to Break and Form Habits," *ACM Trans. Comput.-Hum. Interact.*, vol. 25, no. 3, pp. 1–66, Jun. 2018, doi: 10.1145/3196830.
- [29] Y. Zhu, Y. Long, H. Wang, K. P. Lee, L. Zhang, and S. J. Wang, "Digital Behavior Change Intervention Designs for Habit Formation: Systematic Review," *J. Med. Internet Res.*, vol. 26, p. e54375, May 2024, doi: 10.2196/54375.
- [30] **F. Reinsch, M. L. Sitzberger, M. Kählig, and J. Stark, "Built for sprints, needed for marathons: Mapping the gap between science, users, and implementation in digital habit formation apps," in *Proceedings of the 34th european conference on information systems (ECIS 2026)*, Milan, Italy, 2026.**
- [31] **T. Weimann and J. Stark, "Supporting habit formation for long-term weight loss maintenance with a virtual coach: a research model," in *Wirtschaftsinformatik 2022 proceedings, 2022*, p. 6. [Online]. Available: https://aisel.aisnet.org/wi2022/digital_health/digital_health/6/**
- [32] I. Nahum-Shani *et al.*, "Just-in-Time Adaptive Interventions (JITAs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support," *Ann. Behav. Med.*, vol. 52, no. 6, pp. 446–462, May 2018, doi: 10.1007/s12160-016-9830-8.
- [33] J. Keller, D. Kwasnicka, P. Klaiber, L. Sichert, P. Lally, and L. Fleig, "Habit formation following routine-based versus time-based cue planning: A randomized controlled trial.," *Br. J. Health Psychol.*, vol. 26, no. 3, pp. 807–824, Sep. 2021, doi: 10.1111/bjhp.12504.
- [34] R. Pimm, C. Vandelandotte, R. E. Rhodes, C. Short, M. J. Duncan, and A. L. Rebar, "Cue Consistency Associated with Physical Activity Automaticity and Behavior," *Behav. Med.*, vol. 42, no. 4, pp. 248–253, Oct. 2016, doi: 10.1080/08964289.2015.1017549.
- [35] K. Stawarz, B. Gardner, A. Cox, and A. Blandford, "What influences the selection of contextual cues when starting a new routine behaviour? An exploratory study," *BMC Psychol.*, vol. 8, no. 1, p. 29, Dec. 2020, doi: 10.1186/s40359-020-0394-9.
- [36] B. Gardner *et al.*, "Developing habit-based health behaviour change interventions: twenty-one questions to guide future research," *Psychol. Health*, vol. 38, no. 4, pp. 518–540, Apr. 2023, doi: 10.1080/08870446.2021.2003362.
- [37] B. Verplanken and S. Orbell, "Reflections on past behavior: a self-report index of habit strength," *J. Appl. Soc. Psychol.*, vol. 33, no. 6, pp. 1313–1330, 2003, doi: 10.1111/j.1559-1816.2003.tb01951.x.
- [38] B. Gardner, G.-J. De Bruijn, and P. Lally, "A Systematic Review and Meta-analysis of Applications of the Self-Report Habit Index to Nutrition and Physical Activity Behaviours," *Ann. Behav. Med.*, vol. 42, no. 2, pp. 174–187, Oct. 2011, doi: 10.1007/s12160-011-9282-0.
- [39] A. L. Rebar, S. Elavsky, J. P. Maher, S. E. Doerksen, and D. E. Conroy, "Habits Predict Physical Activity on Days When Intentions Are Weak," *J. Sport Exerc. Psychol.*, vol. 36, no. 2, pp. 157–165, Apr. 2014, doi: 10.1123/jsep.2013-0173.
- [40] R. W. Holland, H. Aarts, and D. Langendam, "Breaking and creating habits on the working floor: A field-experiment on the power of implementation intentions," *J. Exp. Soc. Psychol.*, vol. 42, no. 6, pp. 776–783, Nov. 2006, doi: 10.1016/j.jesp.2005.11.006.
- [41] G. Judah, B. Gardner, and R. Aunger, "Forming a flossing habit: An exploratory study of the psychological determinants of habit formation.," *Br. J. Health Psychol.*, vol. 18, no. 2, pp. 338–353, May 2013, doi: 10.1111/j.2044-8287.2012.02086.x.
- [42] B. Gardner, K. Sheals, J. Wardle, and L. McGowan, "Putting habit into practice, and practice into habit: a process evaluation and exploration of the acceptability of a habit-based dietary behaviour change intervention.," *Int. J. Behav. Nutr. Phys. Act.*, vol. 11, pp. 194–214, Nov. 2014.
- [43] M. Stojanovic, A. Grund, and S. Fries, "Context Stability in Habit Building Increases Automaticity and Goal Attainment," *Front. Psychol.*, vol. 13, p. 883795, Jun. 2022, doi: 10.3389/fpsyg.2022.883795.
- [44] P. Lally and B. Gardner, "Promoting habit formation," *Health Psychol. Rev.*, vol. 7, no. sup1, pp. S137–S158, May 2013, doi: 10.1080/17437199.2011.603640.

- [45] T. L. Webb and P. Sheeran, "Identifying good opportunities to act: Implementation intentions and cue discrimination," *Eur. J. Soc. Psychol.*, vol. 34, no. 4, pp. 407–419, Jul. 2004, doi: 10.1002/ejsp.205.
- [46] M. A. Adriaanse, D. T. D. De Ridder, and J. B. F. De Wit, "Finding the Critical Cue: Implementation Intentions to Change One's Diet Work Best When Tailored to Personally Relevant Reasons for Unhealthy Eating," *Pers. Soc. Psychol. Bull.*, vol. 35, no. 1, pp. 60–71, Jan. 2009, doi: 10.1177/0146167208325612.
- [47] F. Reinsch, T. G. Weimann, and J. Stark, "Tailoring Health: Contextual Variables In Health Recommender Systems," in *Proceedings of the 6th Workshop on Health Recommender Systems co-located with ACM RecSys 2024*, Bari, Italy: CEUR Workshop Proceedings, Oct. 2024. [Online]. Available: https://ceur-ws.org/Vol-3823/1_Reinsch_tailoring_159.pdf
- [48] F. Reinsch, M. Kählig, M. Neubauer, J. Stark, and H. Schlieter, "The App, the Habit, and the Change: Digital Tools for Multi-Domain Behavior Change," in *Proceedings of the 20th International Conference on Information Systems*, 2025, In press.
- [49] K. Gand, J. Stark, H. Schlieter, C. Gißke, and M. Burwitz, "Using Clinical Pathways To Virtual Coach Elderly For Home Rehabilitation Completed Research Paper," in *ICIS 2021 Proceedings*, Dec. 2021.
- [50] H. Schlieter, J. Stark, M. Burwitz, and R. Braun, "Terminology for evolving design artifacts," in *Proceedings of the 14th international conference on wirtschaftsinformatik*, Siegen, Germany, Feb. 2019.
- [51] S. Michie *et al.*, "Representation of behaviour change interventions and their evaluation: Development of the Upper Level of the Behaviour Change Intervention Ontology," *Wellcome Open Res.*, vol. 5, p. 123, Jan. 2021, doi: 10.12688/wellcomeopenres.15902.2.
- [52] F. Reinsch, T. Weimann, and J. Stark, "From Instant Cues to Fading Views: Piloting Just-in-Time and Fading Strategies in Digital Habit Formation," in *AMCIS 2025 Proceedings*, Aug. 2025. [Online]. Available: https://aisel.aisnet.org/amcis2025/health_it/sig_health/5
- [53] M. J. Page *et al.*, "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," *BMJ*, p. n71, Mar. 2021, doi: 10.1136/bmj.n71.
- [54] R. C. Nickerson, U. Varshney, and J. Muntermann, "A method for taxonomy development and its application in information systems," *Eur. J. Inf. Syst.*, vol. 22, no. 3, pp. 336–359, May 2013, doi: 10.1057/ejis.2012.26.
- [55] J. Feine, U. Gnewuch, S. Morana, and A. Maedche, "A Taxonomy of Social Cues for Conversational Agents," *Int. J. Hum.-Comput. Stud.*, vol. 132, pp. 138–161, Dec. 2019, doi: 10.1016/j.ijhcs.2019.07.009.
- [56] H. Ma, A. Wang, R. Pei, and M. Piao, "Effects of habit formation interventions on physical activity habit strength: meta-analysis and meta-regression," *Int. J. Behav. Nutr. Phys. Act.*, vol. 20, no. 1, pp. 1–12, Sep. 2023, doi: 10.1186/s12966-023-01493-3.
- [57] P. Lally, C. H. M. van Jaarsveld, H. W. W. Potts, and J. Wardle, "How are habits formed: Modelling habit formation in the real world," *Eur. J. Soc. Psychol.*, vol. 40, no. 6, pp. 998–1009, 2010, doi: 10.1002/ejsp.674.
- [58] N. Kaushal and R. Rhodes, "Exercise habit formation in new gym members: a longitudinal study," *J. Behav. Med.*, vol. 38, no. 4, pp. 652–663, Aug. 2015, doi: 10.1007/s10865-015-9640-7.
- [59] G. Stadler, G. Oettingen, and P. M. Gollwitzer, "Physical Activity in Women," *Am. J. Prev. Med.*, vol. 36, no. 1, pp. 29–34, Jan. 2009, doi: 10.1016/j.amepre.2008.09.021.
- [60] G. Meyerowitz-Katz, S. Ravi, L. Arnolda, X. Feng, G. Maberly, and T. Astell-Burt, "Rates of Attrition and Dropout in App-Based Interventions for Chronic Disease: Systematic Review and Meta-Analysis," *J. Med. Internet Res.*, vol. 22, no. 9, p. e20283, Sep. 2020, doi: 10.2196/20283.
- [61] D. Schoenfeld, "The asymptotic properties of nonparametric tests for comparing survival distributions," *Biometrika*, vol. 68, no. 1, pp. 316–319, 1981, doi: 10.1093/biomet/68.1.316.
- [62] M. Neubauer, D. Schick, M. Schreiter, J. Stark, T. Eymann, and H. Schlieter, "Bridging digital transformation in public health with digital responsibility," *Electron. Mark.*, vol. 36, no. 1, p. 29, Mar. 2026, doi: 10.1007/s12525-025-00868-7.
- [63] S. Mollborn, E. M. Lawrence, and R. A. Hummer, "A gender framework for understanding health lifestyles," *Soc. Sci. Med.*, vol. 265, p. 113182, Nov. 2020, doi: 10.1016/j.socscimed.2020.113182.
- [64] P. D. Lunn, S. Timmons, C. A. Belton, M. Barjaková, H. Julienne, and C. Lavin, "Motivating social distancing during the COVID-19 pandemic: An online experiment," *Soc. Sci. Med.*, vol. 265, p. 113478, Nov. 2020, doi: 10.1016/j.socscimed.2020.113478.
- [65] J. Meers, J. Stout-Aguilar, and S. Nowakowski, "Sex differences in sleep health," in *Sleep and Health*, Elsevier, 2019, pp. 21–29. doi: 10.1016/B978-0-12-815373-4.00003-4.
- [66] J. Stark and D. Dörfel, "Fast and effective stress relief in women through brainwave-entrainment technology in pandemics," in *ECIS 2021 research papers*, 2021, p. 86. [Online]. Available: https://aisel.aisnet.org/ecis2021_rp/86
- [67] S. Yang *et al.*, "Understanding Inequalities in Mobile Health Utilization Across Phases: Systematic Review and Meta-Analysis," *J. Med. Internet Res.*, vol. 27, pp. e71349–e71349, Aug. 2025, doi: 10.2196/71349.
- [68] J. M. Lipschitz, C. K. Pike, T. P. Hogan, S. A. Murphy, and K. E. Burdick, "The Engagement Problem: a Review of Engagement with Digital Mental Health Interventions and Recommendations for a Path Forward," *Curr. Treat. Options Psychiatry*, vol. 10, no. 3, pp. 119–135, Aug. 2023, doi: 10.1007/s40501-023-00297-3.
- [69] K. Stawarz, M. D. Rodríguez, A. L. Cox, and A. Blandford, "Understanding the use of contextual cues: design implications for medication adherence technologies that support remembering," *Digit. Health*, vol. 2, p. 2055207616678707, Jan. 2016, doi: 10.1177/2055207616678707.

4 Supplementary information on the research context

4.1 Ethical and/or legal aspects of the project

4.1.1 General ethical aspects

The project investigates how contextual cue configurations within implementation intentions shape the transition from intentional to automatic behavior and what design knowledge can be derived to support the sustained formation of health-related habits. Its empirical work involves human participants interacting voluntarily with a digital study platform over an extended period. The research does not target vulnerable populations, involves no deception, no clinical interventions, and no collection of sensitive health data. No physical, psychological, or social risks to participants are anticipated.

Ethics approval for the study was granted by the Ethics Committee of TU Dresden (Ethikkommission an der TU Dresden) on 11 September 2025 (reference: SR-EK-222052025, study title: HabConnect – from Habit to Health). The committee raised no ethical or legal objections to the conduct of the research based on the submitted study protocol, informed consent documents, and data protection concept, and issued its approval pursuant to § 1 Abs. 3 of the TU Dresden Ethics Committee statutes. The approval covers all participant-involving research activities planned within the project and ensures immediate deployment readiness upon project commencement. Should the project extend beyond five years from the date of approval, an interim report will be submitted to the ethics committee as required. In addition, the TU Dresden Data Protection Officer reviewed the Health-Habit-Hub platform infrastructure and issued formal data protection clearance on 28 March 2025 (Az. 0543-025/001), confirming no data protection objections to the conduct of research using this infrastructure.

4.1.2 Descriptions of proposed investigations involving humans, human materials or identifiable data

Selection criteria. Participation is open to adults aged 18 and older with sufficient language fluency who adopt a daily physical-activity behavior as the focus of habit formation. The specific activity is chosen by the participant from a predefined menu of methodologically comparable options (e.g., walking, light jogging, cycling, structured calisthenics, yoga). Persons under 18 are explicitly excluded. In line with the ethics committee's guidance (SR-EK-222052025), any dataset in which a participant indicates an age below 18 will be deleted prior to analysis, even if majority status was confirmed at the point of consent. No further exclusion criteria apply beyond access to the study platform.

Risks and precautions. The study involves no physical interventions, no deception, and no collection of sensitive health data beyond self-reported behavioral frequency and SRHI scores. Target behaviors are everyday health behaviors participants already intend to adopt; no clinical regimen is prescribed. Dropout is self-directed and carries no consequences. All data are pseudonymized at the point of collection via a study ID system, consistent with the data protection clearance (Az. 0543-025/001). Data are processed in full compliance with GDPR requirements and shared only within the project's research group at TU Dresden.

Informed consent. The informed consent procedure was reviewed and approved as part of the ethics committee assessment (SR-EK-222052025). Prior to enrollment, participants are fully informed of the study's purpose, procedures, data handling practices, including that data are shared only within the project's research group at TU Dresden and not passed to any other third parties, their right to withdraw at any time without consequences, and the pseudonymization protocol. Active informed consent is obtained digitally via the platform's onboarding module before any data are collected.

4.1.3 Descriptions of proposed investigations involving experiments on animals

Not applicable.

4.1.4 Descriptions of projects involving genetic resources (or associated traditional knowledge) from a foreign country

Not applicable.

4.1.5 Explanations regarding any possible safety-related aspects

4.1.5.1 "Dual Use Research of Concern"; foreign trade law

Not applicable.

4.1.5.2 Risks in international cooperation

Not applicable.

4.1.6 Considerations on aspects of ecological sustainability in the planning and implementation of the project

The project's ecological footprint is anticipated to be low. The longitudinal field study (WP C) is conducted entirely via a digital platform, avoiding the logistical and emissions burden of laboratory or in-person data collection. Travel is planned exclusively for the dissemination of results at scientific conferences; rail travel will be prioritized for national venues, and virtual participation will be used where feasible and appropriate. Computational resources for platform operation and data analysis draw on TU Dresden's Centre for Information Services and High-Performance Computing (ZIH), which operates under an institutional energy efficiency mandate. The statistical analyses in WP C rely on standard multilevel and survival analysis methods; local analysis work will be carried out on a dedicated research notebook requested under 5.1.2.1.

4.2 Employment status information

Stark, Jeannette; Research Associate, TU Dresden (Research Group Digital Health). Fixed-term, third-party funded position, contract valid until 04/2028. An extension of the employment contract is anticipated and will be requested through the Research Group Digital Health's ongoing third-party funding portfolio; continuation of the applicant's involvement through project completion is institutionally supported.

4.3 First-time proposal data

Stark, Jeannette

4.4 Composition of the project group

Dr. Jeannette Stark, Research Associate, TU Dresden (Research Group Digital Health). Employed on third-party funded projects (not funded through this proposal). Contributes to conceptual development, study design, and supervision.

apl. Prof. Dr. Hannes Schlieter, Professor of Information Systems, TU Dresden (Research Group Digital Health). Funded through institutional resources and third-party funded projects. Contributes to project supervision, strategic alignment, and scientific guidance.

4.5 Researchers in Germany with whom you have agreed to cooperate on this project

None.

4.6 Researchers abroad with whom you have agreed to cooperate on this project

None.

4.7 Researchers with whom you have collaborated scientifically within the past three years

- apl. Prof. Dr. habil. Hannes Schlieter*
- Dr. rer. pol. Martin Burwitz*
- Felix Reinsch*
- Carola Martin (geb. Gißke)*
- Emily Hickmann*
- Maren Kählig*
- Dr. rer. pol. Maria Neubauer*
- Dr. rer. pol. Peggy Richter*
- Marcel Susky*
- Thure Georg Weimann*
- Nick Heidmann*
- Madita Schädlich*
- Rebecca Scheel*
- Prof. Dr. Torsten Eymann, University of Bayreuth
- Prof. Dr. Jeffrey Parsons, Memorial University of Newfoundland
- Doreen Schick, Fraunhofer Institute for Applied Information Technology FIT
- Melina Schreiter, Fraunhofer Institute for Applied Information Technology FIT
- Dr. rer. nat. Stefan Greulich, Jade Hochschule Oldenburg
- Prof. Dr. Alfred Benedikt Brendel, Indiana University, Kelley School of Business
- Prof. Dr. Martin Gersch, FU Berlin
- Prof. Dr. Daniel Fürstenau, FU Berlin
- Prof. Dr. Steffen Greiffenberg, Duale Hochschule Sachsen
- Prof. Dr. Werner Esswein, TU Dresden
- Prof. Dr. Sebastian Pannasch, TU Dresden

* Research Group Digital Health, TU Dresden

4.8 Project-relevant cooperation with commercial enterprises

Not applicable.

4.9 Project-relevant participation in commercial enterprises

Not applicable.

4.10 Scientific equipment

The project relies primarily on the Health-Habit-Hub, a research-focused digital application developed at TU Dresden that has been validated in prior studies and serves as the methodological backbone of the proposed field study. The platform's backend infrastructure (Neo4j graph database, MongoDB event logging) and its hosting environment at TU Dresden are available for the project without additional instrumentation costs. Standard office equipment is available at the applicant's chair. For computationally intensive data analyses in WP C, the Centre

for Information Services and High-Performance Computing (ZIH) at TU Dresden provides access to computing resources through existing institutional arrangements. The only additional equipment required is the high-performance research notebook specified under 5.1.2.1, which serves as the doctoral researcher's primary work instrument for platform development, field study management, and local execution of language models for data processing in compliance with GDPR requirements.

4.11 Other submissions

No funding proposals for this project or for any closely related projects have been submitted to any third-party funding body.

4.12 Other information

The applicant's position is funded through ongoing third-party projects until 04/2028. Beyond this period, continued involvement in the project is supported by the institutional environment and ongoing project portfolio of the Research Group Digital Health at TU Dresden.

5 Requested modules/funds

5.1 Basic Module

5.1.1 Funding for Staff

The proposed project will be carried out over 36 months and requires the support of one full-time doctoral researcher. The work program encompasses a systematic literature review and cue systematization (WP A), platform configuration and pretest (WP B), a large-scale longitudinal field study with continuous enrollment and multilevel data analysis (WP C), and consolidation and dissemination of results (WP D). A full-time position for the entire project duration is essential to ensure methodological rigor and timely completion. The doctoral researcher will be employed at TV-L E13, 100%, for 36 months. Personnel costs are calculated based on DFG standard rates. In addition, the support of a student assistant (TV-L, 40 h/month) for 36 months is requested to assist with technically demanding tasks across multiple work packages, including literature screening in WP A, pretest logistics and participant communication in WP B and WP C, and data preparation in WP C.

Position	Monthly rate	Duration	Total
Doctoral researcher*	6,800 €	36 months	244,800 €
Student Assistant (40 h/month)	825 €	36 months	29,700 €
Total			274,500 €

Table 3: Requested Funding for Staff (* Based on DFG Personnel Rates for 2026)

5.1.2 Direct Project Costs

5.1.2.1 Equipment up to € 10,000, Software and Consumables

Funding for one high-performance notebook (Apple MacBook Pro 16", M5 Max, 128 GB, 4 TB) is requested. The device is required for the local execution of large language models used to structure and store research data in the project's Neo4j graph database. Local execution ensures GDPR compliance by avoiding the transfer of participant data to external cloud services. The M5 Max unified memory architecture enables on-device inference of current-generation language models. The device additionally supports platform development and testing (WP B), field study management (WP C), and data analysis (WP D). The estimated cost is 7.000 € based on the current manufacturer list price.

Item	Calculation	Cost
1 high-performance notebook	7,000 € × 1	7,000 €
Apple Developer Program License	100 € × 3	300 €
Total		7,300 €

5.1.2.2 Travel Expenses

Travel expenses are requested for the doctoral researcher to support dissemination of research findings at major scientific conferences in information systems, digital health, and behavioral science—including venues such as ICIS, ECIS, JMIR-affiliated conferences, and CHI. Two national and one international conference per year are planned for the full three-year project duration.

National Conferences (2 per year × 3 years = 6 total)

Item	Calculation	Cost
Conference fees	650 € × 6	3,900 €
Rail travel	250 € × 6	1,500 €
Accommodation (4 nights/ conference)	150 € × 4 × 6	3,600 €
Per diem (5 days/ conference)	40 € × 5 × 6	1,200 €
<i>Subtotal national</i>		<i>10,200 €</i>

International Conferences (1 per year × 3 years = 3 total)

Item	Calculation	Cost
Conference fees	750 € × 3	2,250 €
Flights	900 € × 3	2,700 €
Accommodation (5 nights/ conference)	200 € × 5 × 3	3,000 €
Per diem (6 days/ conference)	80 € × 6 × 3	1,440 €
<i>Subtotal international</i>		<i>9,390 €</i>
Total		19,590 €

5.1.2.3 Visiting Researchers (excluding Mercator Fellows)

Not requested.

5.1.2.4 Expenses for Laboratory Animals

Not applicable.

5.1.2.5 Other Costs

To recruit participants for the pretest in WP B (20–30 participants, approx. 60-minute session) and to support voluntary participant engagement during the longitudinal study in WP C, a modest compensation budget is requested. Because dropout is a primary outcome measure (H2b), milestone-based compensation would confound behavioral observation by introducing financial incentives to remain enrolled. To avoid this confound, participants receive a one-time upfront payment upon successful completion of onboarding and baseline data provision. This approach compensates participants fairly for their time without artificially influencing the dropout patterns the study aims to observe.

Item	Calculation	Cost
WP B Pretest compensation	30 participants × 10 €	300 €
WP C enrollment compensation (upfront)	1,000 participants × 20 €	20,000 €
Total		20,300 €

5.1.2.6 Project-related Publication Expenses

The project plans two journal publications as described in WP D.2: the cue systematization (targeting a practice-oriented journal such as JMIR or BMC Digital Health) and the field study findings (targeting a high-impact outlet such as MISQ or EJIS). To ensure broad accessibility, open-access publication is intended for both articles. While institutional DEAL agreements cover several target journals, not all planned outlets fall within these agreements. Funding is therefore requested to cover residual article processing charges where institutional agreements do not apply.

Item	Calculation	Cost
Article processing charges	750 € × 3 years	2,250 €
Total		2,250 €

5.1.3 Instrumentation**5.1.3.1 Equipment exceeding € 10,000**

Not requested.

5.1.3.2 Major Instrumentation exceeding € 50,000

Not requested.

5.2 Module Temporary Position for Principal Investigator

Not requested.

5.3 Module Replacement Funding

Not requested.

5.4 Module Temporary Clinician Substitute

Not requested.

5.5 Module Mercator Fellows

Not requested.

5.6 Module Workshop Funding

Not requested.

5.7 Module Public Relations Funding

Not requested.

5.8 Module Standard Allowance for Equity and Diversity

Not requested.